

Research report

Gut microbiota in Parkinson disease in a northern German cohort



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ABSTRACT

Pathologic and epidemiologic studies suggest that Parkinson disease (PD) may in some cases start in the enteric nervous system and spread via the vagal nerve to the brainstem. Mounting evidence suggests that the gut microbiome plays an important role in the communication between gut and brain and that alteration of the gut microbiome is involved in the pathogenesis of numerous diseases, including Parkinson disease. The aim of this study was to determine whether Parkinson disease is associated with qualitative or quantitative changes in the gut microbiome.

We analyzed the gut microbiome in 29 PD cases and 29 age-matched controls by next-generation-sequencing of the 16S rRNA gene and compared diversity indices and bacterial abundances between cases and controls. Alpha diversity measures and the abundance of major phyla did not differ between cases and controls. Beta diversity analyses and analysis on the bacterial family level revealed significant differences between cases and controls for four bacterial families. In keeping with recently published studies, Lactobacillaceae were more abundant in cases. Barnesiellaceae and Enterococcaceae were also more abundant in cases in this study but not in other studies. Larger studies, accounting for drug effects and further functional investigations of the gut microbiome are necessary to delineate the role of the gut microbiome in the pathogenesis of PD.

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1. Introduction

Parkinson disease (PD) is the second most common neurodegenerative disease with a steady increase in prevalence with age. A prevalence peak with approximately 1% affected individuals is reported around the age of 70 (Pringsheim et al., 2014). The cardinal motor symptoms are bradykinesia, rigidity, resting tremor and postural instability (Kalia and Lang, 2015). Non-motor symptoms such as constipation, hyposmia, orthostatic hypotension and urogenital dysfunction are an important part of PD (Chaudhuri et al., 2011). Constipation is consistently associated with an increased risk of PD in epidemiologic studies and often appears years prior to motor symptoms. Based on autopsy studies, Braak and colleagues proposed that PD starts in the enteric nervous system and/or the olfactory bulb (Dickson et al., 2009; Hawkes et al., 2010). The gut microbiome influences the enteric nervous system,

and via the vagal nerve the central nervous system (Ulusoy et al., 2013; Cryan and Dinan, 2012). Complete vagotomy reduces the risk to develop PD (Svensson et al., 2015). This study explores whether PD patients (n = 29) have gut microbiota differences compared to control subjects (n = 29), which might support a role of the gut microbiota in the PD pathophysiology.

2. Results

2.1. Demographics and Clinical data

Table A.1 contains detailed demographic and clinical data. The age distribution in cases (mean: 69.2 yrs., standard deviation (sd): 6.5 yrs) and controls (mean: 69.4 yrs., sd: 6.7 yrs) was nearly identical. The sex ratio differed between cases (23 male, 6 female) and controls (13 male, 16 female) ($p_{\text{Chi}^2} = 0.014$). Constipation was more frequent in patients (20/29) than controls (4/29) ($p_{\text{Chi}^2} = 0.014$), nicotine use was more frequent in controls (8/29) than patients (0/29) ($p_{\text{Chi}^2} = 0.008$) and caffeine consumption was more frequent in controls (23/29) than in patients (8/29).

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($p_{\text{Chi2}} = 0.0002$). Alcohol consumption (yes/no) and eating fermented milk products did not differ between patients and controls. The mean age of onset in the case group was 57.9 yrs. with an sd of 7.1 yrs., the mean UPDRS part III score was 21 and 6 PD cases were treated with deep brain stimulation (DBS).

2.2. Intestinal microbiota

We detected no significant differences between cases and controls in the abundances of major phyla and also no significant differences in alpha diversity (species richness of a group) measures (Simpson, Shannon, Chao1, Phylogenetic distance; Mann-Whitney U test, $p > 0.05$). Alpha diversity indices were also not dependent on sex and age (ANOVA, $p > 0.05$ for each factor). We assessed potential community-level differences between samples using beta diversity analyses, which consider the actual bacterial community composition of each group. The unweighted UniFrac dissimilarity significantly differs between cases and controls (*adonis*: unweighted UniFrac $R^2 = 0.0241$, $p = 0.0491$), but not between sexes ($R^2 = 0.01621$, $p = 0.553$). Next, we analyzed bacterial family-level abundances to identify the bacterial families responsible for the differences in the beta diversity. The ten most abundant bacterial groups accounted for ~80% of the total bacterial count (Fig. A.1). Using a 1000-fold Monte-Carlo simulation Wilcoxon test the abundance of four bacterial families differed at first sight significantly between cases and controls. However, quasi-Poisson regression analysis with bacterial counts as dependent variable and manual backwards selection of case/control status, age, sex, constipation, nicotine use, caffeine consumption, alcohol consumption, use of fermented milk products, cardiovascular disease and sequencing batch as independent variables for these families showed that sex and not disease status was the significant determinant of Lachnospiraceae count ($p = 0.0001$). Case/control status was a significant determinant ($p < 0.05$) of bacterial abundance for Barnesiellaceae and Enterococcaceae and a suggestive determinant ($p < 0.1$) for Lactobacillaceae (Tables 1, 2 and Fig. 1). Since a previous large study explicitly mentioned that COMT-inhibitors had an influence on bacterial counts (Hill-Burns et al., 2017), we performed an analysis without the four patients taking COMT-inhibitors, which did not substantially change the results (Table 1).

2.3. Comparison to previous studies

Four previous studies that are in principle comparable addressed differences in the microbiota between PD patients and

controls (Hill-Burns et al., 2017; Hasegawa et al., 2015; Keshavarzian et al., 2015; Scheperjans et al., 2015). The studies are not directly comparable due to differences in wet-lab techniques as well as in statistical analysis strategies. Therefore, we limit the comparison to bacterial families showing significant differences between cases and controls in at least one study (Table 1). Bacteria of the Lactobacillaceae group were identified to be more abundant in cases than controls in four out of five studies, including this one (Hill-Burns et al., 2017; Hasegawa et al., 2015; Scheperjans et al., 2015). Verrucomicrobiaceae were more abundant in cases than controls in two studies but not in this one (Hill-Burns et al., 2017; Keshavarzian et al., 2015). The distribution of bacterial families identified in previous studies in our study population is shown in Fig. A.2. We did not include another recently published study in the comparison because the bacterial nomenclature used is not compatible with our study (Unger et al., 2016).

2.4. ROC analysis

To explore the potential usefulness of microbiota analysis as a biomarker for PD we performed a ROC analysis. The area under the curve (AUC) for disease status with Lactobacillaceae abundance as predictor was 0.68 (95% CI: 0.58–0.79) (Fig. A.3). The same analysis with Lactobacillaceae, Barnesiellaceae and Enterococcaceae abundance as predictors resulted in an AUC of 0.83 (95% CI: 0.73–0.94; Fig. A.4), indicating that it might be possible to derive a PD biomarker from intestinal microbiota analysis.

3. Discussion

We found a significant increase of three bacterial families in PD patients compared to age matched controls (Table 1). A significant increase of Lactobacillaceae in PD patients is in accordance with three previous studies, while the fourth study provides no information on Lactobacillaceae (Hill-Burns et al., 2017; Keshavarzian et al., 2015; Scheperjans et al., 2015). In our study, PD patients did not report probiotic use or other dietary habits explaining the difference in Lactobacillaceae abundance. We also found higher abundances of Enterococcaceae and Barnesiellaceae in the case group compared to the control group. ROC analysis showed that microbiota might have a predictive value for PD. However, it should be kept in mind that the AUC in the discovery stage is in nearly all cases overestimated, stressing the need for replication in independent samples. The main weakness of our study is the low number of participants and the imbalanced gender distribu-

Table 1
Comparison of p-values of bacteria associated with PD in this study or in previous studies. Significant p-values (≤ 0.05) in bold print, in brackets: without patients taking COMT-inhibitors. Due to different sequencing approaches and statistical methods, the p-values are not directly comparable. +bacterial count higher in cases, –bacterial count higher in controls, =bacterial count in patients and controls nearly identical.

Bacterial family	This Study	Hill-Burns et al. (2017)	Scheperjans et al. (2015)	Keshavarzian et al. (2015)	Hasegawa et al. (2015)
Statistical method	Permut.	K-W, ANCOM	Permutat.	FDR-BH	MWU
Lactobacillaceae	0.002+ (0.014)	0.0000002+	0.004+	–	0.003+ (Lactobacillus)
Barnesiellaceae	0.007+ (0.011)	–	–	–	–
Enterococcaceae	0.005+ (0.004)	–	–	–	0.082+ (Enterococcus)
Prevotellaceae	0.936–	–	0.001–	0.88=	0.28 (Prevotella)
Verrucomicrobiaceae	0.444+	0.0004+	0.014+	0.05+	–
Bradyrhizobiaceae	0.581–	–	0.021+	–	–
Ruminococcaceae	0.309–	–	0.029+	0.31+	–
Bacteroidaceae	0.364+	–	–	0.05+	–
Bifidobacteriaceae	0.142+	0.000002+	–	0.2	0.47
Tissierellaceae	0.699+	0.00000008+	–	–	–
Christensenellaceae	0.320=	0.000000004+	–	–	–
Pasteurellaceae	–	0.00003–	–	–	–

Table 2

Quasi-Poisson regression with stepwise manual backward selection p-values for bacterial families as dependent variable and case/control status, sex, age, constipation, nicotine, alcohol, caffeine, fermented milk products and cardiovascular disease as independent variables. Variables which did not yield significant or suggestive p-values for any of the bacterial species shown are omitted from the table. Significant and suggestive p-values (<0.1) in bold print.

Bacteria	Case/Control	Sex	Constipation	Alcohol
Lactobacillaceae	0.073	–	–	–
Barnesiellaceae	0.012	–	–	–
Enterococcaceae	0.034	0.097	0.048	0.0004

tion. In common with other published studies we cannot rule out that the findings are due to drug treatment of PD since all patients were treated. This would require a second control group taking a similar medication as PD patients, e.g. patients with restless legs syndrome or drug naïve patients. A strength of our study is that cases were matched to controls who share the same direct environment and diet. However, additional weaknesses of the study are the lack of detailed data on previous dietary habits and related factors and the high amount of cardiovascular comorbidity in the control group. Both the patients and the controls reported to be omnivores with a conventional diverse diet. Our study, in conjunction with the already published studies, shows that a large scale, well controlled study including control groups taking dopaminergic drugs and drug naïve PD patients is needed to obtain detailed, robust data about changes in the gut microbiota in PD. To allow replication and assess the potential of gut microbiota composition

as a biomarker of PD the study needs to include at least two independent case as well as control groups.

4. Methods and materials

4.1. Subjects

29 PD cases and 29 controls were included in this study. A neurodegenerative disease was ruled out in controls. All participants were recruited in the Dept. of Neurology, Kiel University. The study was approved by the local ethics committee (ethical approval: D421/14) and written informed consent was obtained from all participants.

4.2. Clinical data

Cases (n = 40) and controls (n = 42) from our clinic patients were initially contacted and informed about the study by a letter asking for their participation. All PD cases had been diagnosed by a movement disorder specialist in our clinic according to the UK Parkinson's Disease Society Brain Bank Clinical Diagnostic Criteria (Hughes et al., 1992). In a second stage, cases (n = 29) and controls (n = 29) interested in the study underwent an interview. Antibiotic use within the last three months, as well as gastrointestinal comorbidities were excluded for all participants. Participants were clearly instructed how to collect the fecal samples either at home or at the Department of Neurology. Unified Parkinson's Disease Rating Scale (UPDRS) data were available for all PD cases (Scale,

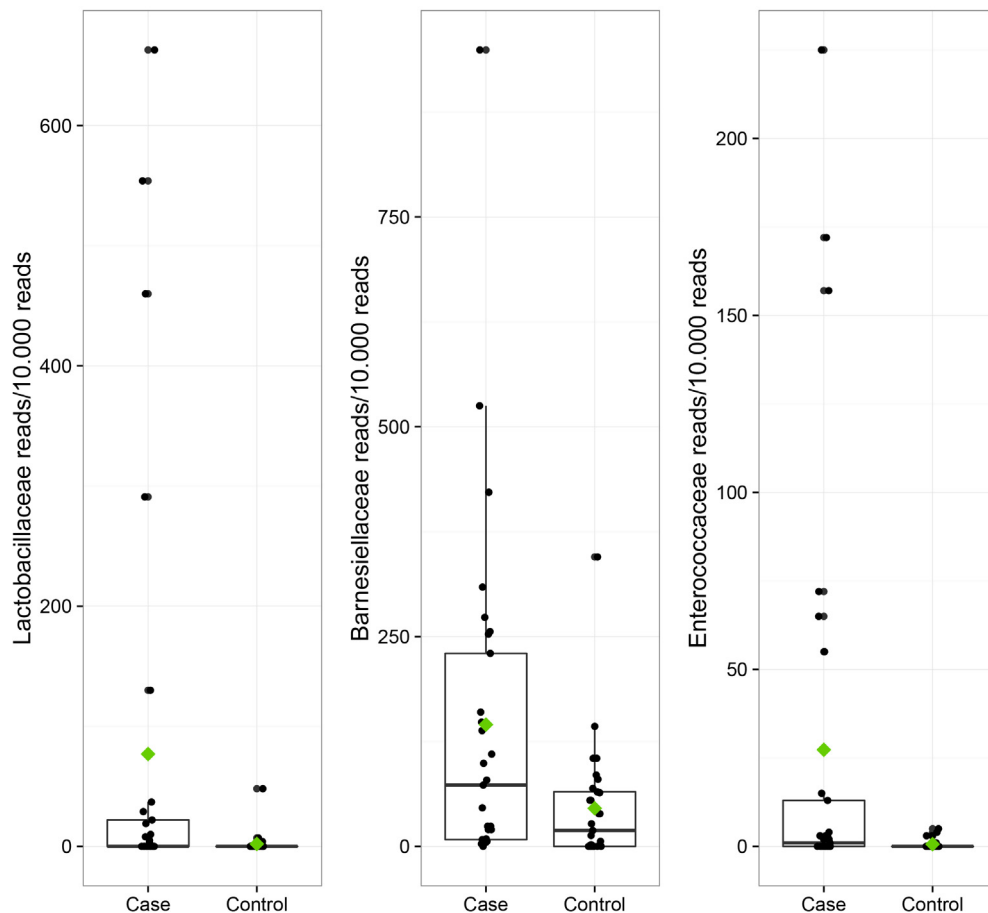


Fig. 1. Distribution of bacterial counts for PD associated bacterial families. Boxplots and scatterplots of the bacterial counts for PD associated bacterial families. Line in the box – median, green rhomb – mean, dots – bacterial counts in single cases/controls.

2003). Additional clinical data including medication were obtained from case records. Signs of PD or prodromal PD in controls were ruled out by examination and questions regarding to smelling, REM sleep disorders, constipation and depression.

4.3. Fecal microbiota

Native stool samples were collected mostly in the hospital and in some cases shortly after discharge at home and transferred to the Department of Neurology in less than 48 h. The fecal sample was stored without preservation solution at room temperature during shipment. As soon as the fecal sample reached our laboratory it was aliquoted and stored at -80°C . We extracted bacterial DNA using the PowerSoil Kit (MoBio, Carlsbad, CA). We amplified parts of the 16S rRNA using barcoded primers flanking the V1 and V2 hypervariable regions and sequenced in two batches using the Illumina MiSeq v2 chemistry with 5% PhiX DNA added to the library. Samples were sequenced using 300 bp paired-end reads. On average, about 152,000 ($\pm 73,000$; median 156,455) read pairs were sequenced per sample.

4.4. Raw data preprocessing

Raw data were processed using USEARCH version 7.0.1090. Overlapping reads were merged into contigs and chimeras were removed using the RDP Gold database version 9 as reference (Edgar, 2010). We performed taxonomic classification using Mothur version 1.34.2 (reference database: GreenGenes version 13_5) with 80% bootstrap support (Schloss et al., 2009). To identify Operational taxonomic units (OTUs), we used 10,000 contigs per individual (97% similarity threshold). In total, 16,991 different OTUs were identified across the 58 samples. OTUs present in only one contig in a single individual (singletons) were removed. 5556 OTUs remained for further analyses. We constructed a phylogenetic tree using FastTree version 2.1.9 with a generalized time-reversible (GTR) substitution model and the gamma option to rescale branch length.

4.5. Statistical analyses

All statistical analyses were performed using R version 3.2.1. Alpha (Chao1, Shannon, Simpson)- and beta (unweighted UniFrac) diversity indices were calculated using the *vegan* package and phylogenetic distances (alpha diversity) were computed according to the method of Faith (Oksanen, 2011; Faith, 1992). We analyzed differences in bacterial family counts between cases and controls using a 1000-fold Monte-Carlo simulation Wilcoxon test. Bacterial abundance data showed strong overdispersion. Therefore, we employed a Quasi-Poisson regression model and manual backward elimination with a $p_{\text{exclusion}} > 0.1$, the dependent variable bacterial family count and the independent variables case/control status, age, sex, constipation, nicotine use, caffeine consumption, alcohol consumption, use of fermented milk products, cardiovascular disease and sequencing batch. We defined “cardiovascular disease” as either: arterial hypertension, history of cardiovascular disease, history of stroke, atrial fibrillation. We used the package “pROC” for Receiver operating characteristic (ROC) analysis.

5. Web Resources – URLs in this manuscript

Coin (version 1.1–2) <http://coin.r-forge.r-project.org>
 FastTree (version 2.1.4) <http://www.microbesonline.org/fasttree/>
 GreenGene database (version 13_5) <http://greengenes.lbl.gov/cgi-bin/nph-index.cgi>

MOTHUR (version 1.34.2) <http://www.mothur.org/>
 R (version 3.2.1) <https://cran.r-project.org/bin/windows/base/old/3.2.1/pROC> (version 1.8) <https://cran.r-project.org/web/packages/pROC/index.html>
 USEARCH (version 7.0.1090) <http://www.drive5.com/usearch/download.html>
 Vegan (version 2.2–1) <https://cran.r-project.org/web/packages/vegan/index.html>

Author contributions

Conception, organization and execution of the research project was done by FH, SHM, SK, KEZ, NGM, GD, JFB and GK. AK and GK performed the data processing and statistical analysis. FH, GK and AK and wrote the manuscript with input from the co-authors.

Competing financial interests

The authors declare no competing financial interests.

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Relevant conflicts of interest: Nothing to report. We thank all patients and controls who contributed to this study and Katja Cloppenburg-Schmidt from the Institute for Experimental Medicine, Christian-Albrechts-University of Kiel, Germany for technical assistance.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.brainres.2017.04.019>.

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